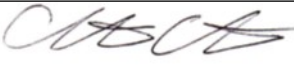


Guideline Title: Pediatric Altered Mental Status (Non-Traumatic)

Guideline Number: GUID-3203-PED003

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Identify and correct immediately reversible causes of altered mental status (AMS) and provide safe and expeditious transport to the closest appropriate facility.
- II. **DEFINITIONS:**
- III. **DEPARTMENT GUIDELINE:**
 - a. Obtain good assessment: Assess airway, breathing, and circulation as per *GUID3203-PED001 – Pediatric General Management*. The AMS patient presents with an altered level of consciousness that does not correspond with their baseline behavior. They may display inappropriate responses to verbal & physical stimuli. Consider HPI, as it may provide indicators as to the underlying cause.
 - b. Consider possible causes
 - i. **Toxic ingestion:** Question caregivers/referring staff about possibility of ingestion, access to substances. Observe for specific toxidromes, including narcotic ingestion (pinpoint pupils, hypoventilation with AMS). Refer to *GUID3203-M019 - Overdose, Poisoning, & Toxic Exposure*.
 - ii. **Hypoxia:** Observe for airway patency, work of breathing (grunting, retractions, nasal flaring), tachypnea, tachycardia or bradycardia (late sign), cyanosis. ABC's, pulse oximetry, or ETCO2 sampling may be useful. Providers must rapidly identify the need for ventilatory support based upon the findings discovered by largely clinical observation in the initial assessment.
 - iii. **Hypovolemia:** Assess for signs of volume deficit including tachycardia, tachypnea, pallor, dry and/or pale mucous membranes, sunken fontanel, sunken eyes, poor skin turgor, and decreased urine output (indwelling urinary catheter, or number of wet diapers in a given time period). Low BP is a *late sign* and should not be used to disregard the diagnosis of hypovolemia in children.
 - iv. **Meningitis/meningoencephalitis:** Observe for fever, nuchal rigidity, petechial rash (non-blanching reddish-purple areas), and/or bulging fontanel.
 - v. **Sepsis:** Assess for history of recent illness, suspected source, or physical portal of entry (wound, umbilical stump, etc.). Clinical signs of sepsis include fever, hypothermia, mottling of skin, and signs of hypovolemia. Review available lab work for increased WBC's, left shift, (+) cultures.
 - vi. **Hypoperfusion of cardiac etiology:** Assess patient for heart rate and regularity. Auscultate anterior chest for murmurs or adventitious cardiac sounds. Palpate brachial and femoral pulses for equality (coarctation of the aorta may present with bounding upper extremity pulses and diminished/absent femorals). In cyanotic heart defects, cyanosis will persist despite appropriate oxygen administration and ventilation. Review EKG for dysrhythmias.

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- vii. **Neurotrauma:** Illicit history from caregivers/referring staff about recent trauma. Assess for head injuries & palpate head for depressed areas or hematomas. Observe for otorrhea, rhinorrhea, bulging fontanelles, or other evidence of basilar skull fractures. Spinal immobilization is appropriate if neurotrauma is suspected.
- viii. **Seizure:** Seizures have numerous presentations. Partial seizures can include absence episodes, abnormal vocalizations, or focal motor including head turning, eye deviation, tonic contractions, lip smacking, pedaling motions, etc. without or with a loss of consciousness (partial complex seizures) and also may secondarily generalize into tonic/clonic activity.
- ix. **Metabolic Issues:**
 - 1. **Diabetic Ketoacidosis (DKA):** Assess initial glucose level. The diagnosis of type I diabetes is often made when the patient presents with DKA. Extremely low ETCO₂ in a spontaneously breathing patient is a sign of severe acidosis/DKA. Refer to GUID3203-PED008 - *Pediatric Diabetic Ketoacidosis* for additional pathophysiology and assessment parameters.
 - 2. **Electrolyte abnormality:** Evaluate completed lab work, if available. Sodium imbalances, result in significant alterations in mental status.
- x. **Intussusception:** Telescoping of the bowel into itself can occur at any location, most common is ileocecal. The most common presentation is acute discomfort followed by lethargy. The classic finding of “currant jelly stools” is a late sign indicating infarction of the bowel.
- xi. **Stroke and intracerebral hemorrhage:** Pediatric ischemic strokes and intra-cranial hemorrhages are rare. Careful history taking may provide clues to the etiology of the presenting problem. Suspect ischemic stroke if symptoms consistent (unilateral deficits, AMS). Prevalence of intra-cranial abnormalities is increased in those with a history of a hematologic disorder (e.g., sickle cell disease) or a compromised clotting cascade (vonWillebrand’s disease & hemophilia). There is also an increased prevalence in infants born at home who did not receive postpartum vitamin K.
- c. Assess and manage airway, breathing and circulation.
- d. Initiate cardiac monitoring and follow pulse oximetry.
- e. Establish an IV/IO.
- f. If volume depletion is suspected, **0.9% NS 20 ml/kg IV** per GUID3203-PED009 - *Pediatric Hypotension*.
- g. If the blood sugar level is less than 60 mg/dL, treat per GUID3203-M021 - *Hypoglycemia*.
- h. If an opioid drug overdose is suspected, administer
 - i. If > 20 kg: **Naloxone 0.1 mg/kg/dose IV/IN/IM/SC** repeat every 2-3 min PRN, max single dose 2 mg.
 - ii. If < 20 kg: **Naloxone 0.01-0.1 mg/kg/dose IV/IN/IM/SC** repeat every 2-3 min PRN, max single dose 2 mg.
- i. If sepsis is suspected, refer to the GUID3203-PED016 - *Pediatric Sepsis/Meningitis*.
- IV. **DOCUMENTATION:** EMS Charts
- V. **REFERENCES:**